

Node.js is a JavaScript runtime that runs outside the browser.

Node.js uses the V8 JavaScript engine.

Node.js is commonly used for APIs, web servers, command-line tools, and real-time applications.

Node.js uses an event loop and non-blocking I/O for handling many operations efficiently.

Asynchronous code can use callbacks, Promises, or `async` and `await`.

The `npm` package manager installs and manages JavaScript dependencies.

`package.json` describes a Node.js project's scripts, dependencies, and metadata.

The `fs` module reads and writes files, while the `path` module creates safe file paths.

Express is a web framework commonly used with Node.js.

An Express route connects an HTTP method and URL to a handler function.

Middleware is a function that can inspect a request and response during a request cycle.

Middleware can authenticate users, log requests, or handle errors.

A route handler sends a response when a request matches a route.

`req` contains request information and `res` is used to send a response.

`next` passes control to the next middleware function.

Express can parse JSON request bodies with `express.json()`.

HTTP GET reads data, POST creates data, PUT or PATCH updates data, and DELETE removes data.

Environment variables keep configuration such as ports and secrets outside source code.

Node.js can connect to databases through database drivers or application libraries.

CORS controls which browser origins may call a server.